

KHWAJA YUNUS ALI UNIVERSITY

School of Science and Engineering
Department of Computer Science and Engineering

Spring-2022 Final Examination

Program: B.Sc. in CSE (Batch – 12th) 1st Year 2nd Semester

Course Title: Physics - II

Course Code: PHY 1201

PART-A: MCQ

Time: 10 Minutes

Total Marks: 10x1=10

Student's ID:

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Date:

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 Invigilator's Signature:

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(These are single best type of questions. Each of the following questions or statements has five suggested answers or completions. Put tick mark (✓) on the best answer. There is no counter marking.)

1. Which is essential for interference?
 - A) Two different color lights.
 - B) The movement of the two waves should be opposite.
 - C) The light sources should be far apart.
 - D) The light sources should be coherent.
 - E) The phase difference should have to be $\frac{\pi}{2}$.
2. Corpuscular theory of light given by _____.
 - A) Newton
 - B) Maxwell
 - C) Max Plank
 - D) Huygen
 - E) Fermat
3. Speed of the electromagnetic wave depends on _____ of the medium.
 - A) elasticity
 - B) molecular structure
 - C) radioactivity
 - D) density
 - E) magnetic permeability
4. The phase difference between two points on a wave is π , what is the path difference between them?
 - A) λ
 - B) $\frac{\lambda}{2}$
 - C) $\frac{\lambda}{4}$
 - D) $\frac{3\lambda}{2}$
 - E) $\frac{3\lambda}{4}$
5. What is the Brewster angle for air to glass transition? (Refractive index of glass = 1.5)
 - A) 36.31°
 - B) 46.31°
 - C) 56.31°
 - D) 66.31°
 - E) 76.31°

P.T.O.

6. What will be the angle between the reflected ray and refracted ray when the light ray incident at a polarizing angle to a surface?
- A) 90°
 - B) 60°
 - C) 45°
 - D) 30°
 - E) 0°
7. The radius of the hydrogen nucleus is _____ .
- A) 0.53 m
 - B) 1.2×10^{-15} m
 - C) 3.2×10^{-15} m
 - D) 13.6×10^{-15} m
 - E) 17.6×10^{-15} m
8. Half-life of a radioactive substance is 10 days. How long will it take to decrease to 25% of it?
- A) 5 days
 - B) 10 days
 - C) 15 days
 - D) 20 days
 - E) 25 days
9. What is the unit of the decay constant?
- A) m
 - B) kg
 - C) J
 - D) Sec
 - E) A
10. A space shuttle crosses the football ground of KYAU along of its length at speed of $0.6c$. The football ground is 100m long and 50m wide. What will be width of the football ground according to the measurement by a foreign astronaut in the space shuttle?
- A) 100m
 - B) 80m
 - C) 50m
 - D) 40m
 - E) 30m

Signature of Examiner

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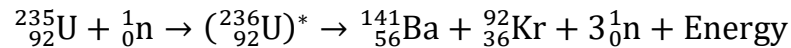
PART-B: SAQ

Time: 1 Hour and 50 Minutes

Total Marks: 5x6=30

INSTRUCTIONS: Answer question no. 01 (one) and any 04 (four) from the rest. Marks are shown by side.

1. See the stem and answer the questions that follows:



Where, $M({}_{92}^{235}\text{U}) = 235.0439 \text{ amu}$; $M({}_{56}^{141}\text{Ba}) = 140.9139 \text{ amu}$;

$M({}_{36}^{92}\text{Kr}) = 91.8973 \text{ amu}$; $M({}_0^1\text{n}) = 1.0087 \text{ amu}$

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|-------|---|---|
| a) | Give a list of the main components of the nuclear reactor. | 1 |
| b) | Calculate the value of the energy of above stem. | 2 |
| c) | Interpret the process of expansion of the above reaction in the reactor and specify a controlling mechanism for that expansion. | 3 |
| 2. a) | Define stopping potential. | 1 |
| b) | How are photoelectrons generated? | 2 |
| c) | UV light of wavelength 350 nm and intensity 1.00 w/m ² is directed at a potassium surface. Find the maximum K.E. of photoelectrons. (Work function of Potassium is 2.2 eV) | 3 |
| 3. a) | Distinguish the inertial and non-inertial frame of references. | 2 |
| b) | Is it possible that the mass of an electron is equal to the mass of a proton? Analyze mathematically. | 4 |
| 4. a) | Define wave front. | 1 |
| b) | Illustrate the light interference that occurs in a thin film. | 2 |
| c) | A thin sheet of Mica ($\mu = 1.58$) is used to cover one slit of the double-slit arrangement. The central point on the screen is now occupied by what to be the seventh bright fringe. If the wavelength of the light used is $5500\text{\AA}$. What is the thickness of the Mica sheet? | 3 |
| 5. a) | What is the dispersive power of grating? | 1 |
| b) | Draw the intensity distribution curve for diffraction of light. | 2 |
| c) | How many orders will be visible (Max. observable angle is 30°) if the wavelength of incident radiation be $5000\text{\AA}$. and the number of lines on the grating be 14000 per inch? | 3 |
| 6. a) | State Malus' Law. | 1 |
| b) | Write some methods of polarization of light. | 2 |
| c) | If the refractive index of a polarizer is 1.9218. What will be the polarization angle and angle of refraction? | 3 |